DANG QUOC THINH

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SUMMARY

Final-year Computer Science student at UIT specializing in Artificial Intelligence, Deep Learning, and Computer Vision. Practical experience building and deploying end-to-end AI systems, including multimodal vector retrieval, medical image classification, and conversational AI voice agents with RAG workflows. Proficient in PyTorch, Transformers, LLM tool calling, and containerized deployment with Docker and Linux.

WORK EXPERIENCE

Intern AI Engineer, VOS Advisory **Jan 2026 – Jul 2026**

- Designed and developed data processing workflows and AI Voice Agent pipelines integrating STT, LLM, RAG, tool calling, and TTS for real-time conversational AI applications.
- Built and delivered backend microservices and APIs using Python, SQL, Docker, Linux, and Git to support scalable AI agent deployment.

EDUCATION

Bachelor of Computer Science **Jul 2023 – Oct 2026 (expected)**

University of Information Technology

- Academic performance: Good (GPA **8.4**/10).

CERTIFICATIONS

- **NVIDIA Applications of AI for Anomaly Detection:** Link
- **VSTEP B2 (Equivalent to IELTS 6.0)**

TECHNICAL SKILLS

- **Deep Learning & CV:** PyTorch | ConvNeXtV2 | Focal Self-Attention | OpenCV | OpenCLIP | Scikit-learn
- **Generative AI & Agents:** LLMs | RAG | AI Voice Agents (STT/TTS) | Tool Calling | Milvus (Vector DB)
- **Reinforcement Learning:** Deep RL (PPO, DDPG, A2C) | Actor-Critic | Custom Gym Environments | MDP Formulation
- **Programming Languages:** Python | C++ | SQL | JavaScript | TypeScript
- **Backend & Deployment:** FastAPI | Docker | Linux | Git | Redis | AWS
- **Databases & Data:** PostgreSQL | MySQL | Pandas | NumPy | Matplotlib | Seaborn

ADDITIONAL SKILLS

- Having experience in working with foreign team members
- Familiar with agile/sprint working environments.
- Able to quickly learn and apply new technologies and modern development tools.

PROJECTS

Skin Cancer Detection Using ConvNeXtV2 and Focal Self-Attention **Mar 2026 – May 2026**

Role: Machine Learning Developer | Member: 3

- **Hybrid Model Architecture:** Re-implemented and evaluated a hybrid deep learning model combining ConvNeXtV2 (local feature extraction) and Focal Self-Attention (global context modeling) to classify 8 skin lesion categories.
- **Data Preprocessing & EDA:** Handled medical image preprocessing, strict 80/10/10 chronological splits, and class balance analysis on **25,331 images** from the ISIC 2019 dataset; applied CutMix, MixUp, and Elastic Deformation.
- **Training & Constraint Analysis:** Conducted training on an NVIDIA Tesla T4 GPU (16GB VRAM) for 40 epochs; stabilized loss curves and systematically analyzed severe class imbalance and hardware-induced underfitting.

Technologies: Python, PyTorch, Scikit-learn, Matplotlib, Seaborn, Git, Linux.

Automated Stock Trading Using Deep Reinforcement Learning **Mar 2026 – May 2026**

Role: Machine Learning Developer | Member: 1

- **MDP & Market Simulation:** Formulated a custom OpenAI Gym environment with a **91-dimensional state space** (portfolio cash, 15 stock prices, technical indicators: MACD, RSI, CCI, ADX) and a 15-dim continuous action space, integrating realistic HOSE market constraints (T+3 settlement, $\pm 7\%$ price collars, 0.1% transaction fees).
- **Actor-Critic & Ensemble Strategy:** Trained and benchmarked PPO, A2C, and DDPG agents using quarterly rolling windows; implemented an ensemble selection policy and a **Turbulence Index** risk circuit breaker to liquidate positions into cash during extreme market volatility.
- **Empirical Performance & Risk Analysis:** Backtested on multi-year out-of-sample data (2021–2025); DDPG achieved **49.66% cumulative return** (11.77% annualized), a **0.6432 Sharpe ratio**, and reduced maximum drawdown to **-16.57%**.

Technologies: Python, TensorFlow, Gym, DDPG, PPO, A2C, Pandas, NumPy, Matplotlib.

Multimodal Video Retrieval System **Sep 2025 – Dec 2025**

Role: Backend & AI Developer | Member: 4

- **Semantic Multimodal Search:** Built an AI retrieval pipeline for text, image, audio, and video, processing **10,000+ multimedia items** through automated embedding and vector-search pipelines.
- **Inference & Optimization:** Developed FastAPI inference services using OpenCLIP and Milvus, achieving ~**150 ms average query latency** under 50 concurrent requests in load testing.
- **Containerization:** Containerized the end-to-end vector retrieval pipeline with Docker for scalable service deployment.

Technologies: Python, FastAPI, OpenCLIP, Milvus, OpenCV, Docker.

Sentiment Analysis using Fine-tuned RoBERTa **Apr 2025 – Jun 2025**

Role: Full Pipeline Developer | Member: 1

- **Transformer Fine-Tuning:** Fine-tuned a RoBERTa-based NLP framework for multi-class sentiment analysis on **20,000+ labeled samples**, optimizing deep learning parameters using PyTorch and Hugging Face Transformers.
- **Data Pipeline & Balancing:** Built automated text preprocessing and tokenization pipelines, implementing dataset balancing techniques to resolve label distribution variance.
- **Benchmarking & Serving:** Benchmarked against baseline estimators (Logistic Regression, SVM), achieving ~**87% weighted F1-score** (+6% over baseline), and containerized API serving using FastAPI and Docker.

Technologies: Python, PyTorch, Hugging Face Transformers, Scikit-learn, FastAPI, Docker.